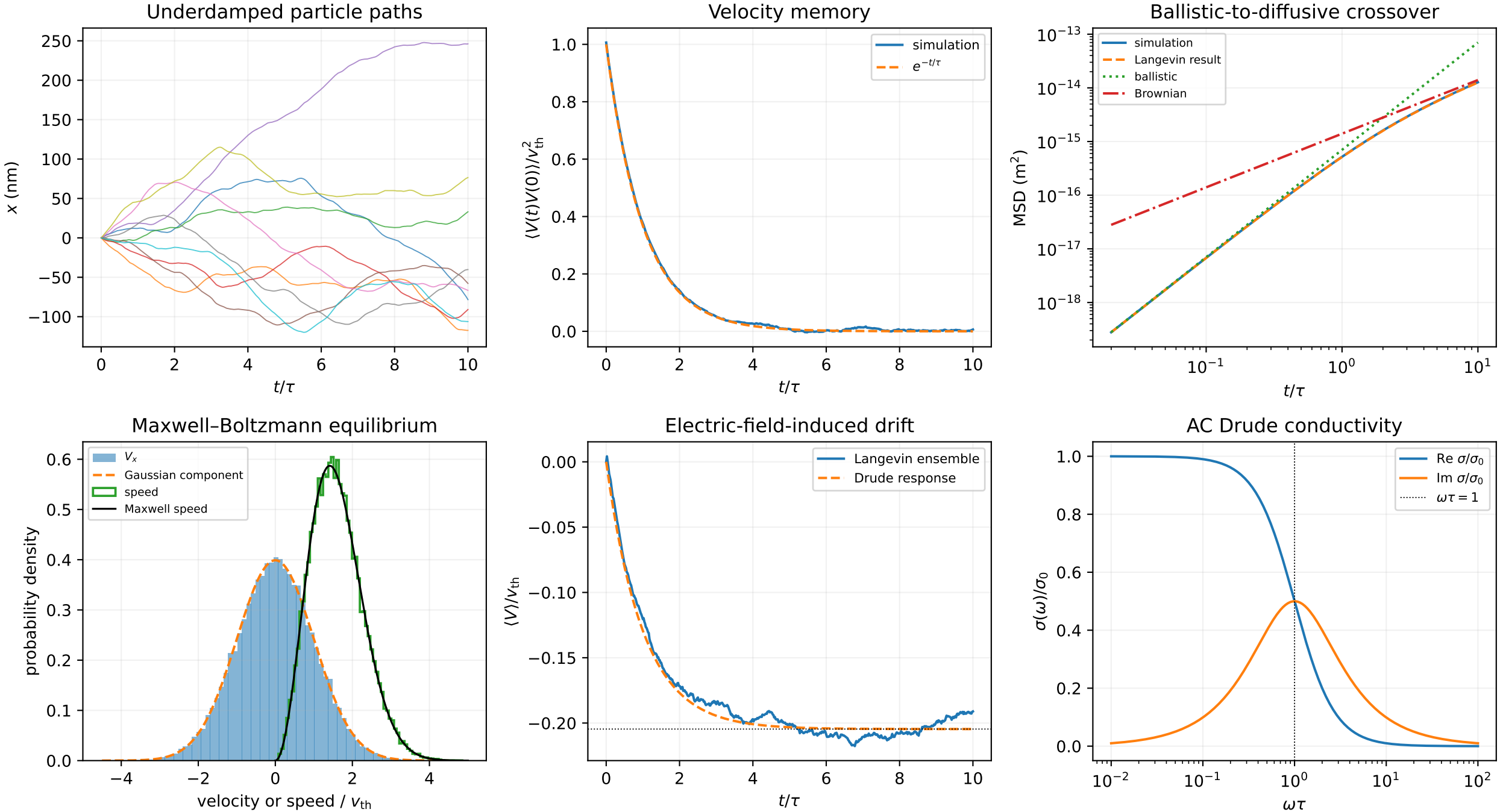

Underdamped particle paths
x (nm)
t/τ
Velocity memory
⟨V(t)V(0)⟩/v²th
simulation
e^{−t/τ}
t/τ
Ballistic-to-diffusive crossover
MSD (m²)
simulation
Langevin result
ballistic
Brownian
t/τ
Maxwell–Boltzmann equilibrium
probability density
V_x
Gaussian component
speed
Maxwell speed
velocity or speed / v_th
Electric-field-induced drift
⟨V⟩/v_th
Langevin ensemble
Drude response
t/τ
AC Drude conductivity
σ(ω)/σ₀
Re σ/σ₀
Im σ/σ₀
ωτ = 1
ωτ